

1. Custom Exception Class

java

```
// Custom exception for insufficient funds
class InsufficientFundsException extends Exception {
    public InsufficientFundsException(String message) {
        super(message); // Pass message to parent Exception class
    }
}
```

2. BankAccount Class

java

```
class BankAccount {
    private String accountHolder;
    private double balance;

    // Constructor
    public BankAccount(String accountHolder, double initialBalance) {
        this.accountHolder = accountHolder;
        this.balance = initialBalance;
    }

    // Deposit method
    public void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            System.out.println("Deposited: $" + amount);
            System.out.println("New Balance: $" + balance);
        } else {
            System.out.println("Deposit amount must be positive!");
        }
    }

    // Withdraw method - throws custom exception
    public void withdraw(double amount) throws InsufficientFundsException {
        if (amount > balance) {
            // Throw custom exception when balance is insufficient
        }
    }
}
```

```

        throw new InsufficientFundsException(
            "Insufficient funds! Available balance: $" + balance +
            ", Requested: $" + amount
        );
    }

    if (amount <= 0) {
        System.out.println("Withdrawal amount must be positive!");
        return;
    }

    balance -= amount;
    System.out.println("Withdrawn: $" + amount);
    System.out.println("Remaining Balance: $" + balance);
}

// Check balance
public void checkBalance() {
    System.out.println("Account Holder: " + accountHolder);
    System.out.println("Current Balance: $" + balance);
}
}

```

3. Main Application Class

java

```

public class BankApplication {
    public static void main(String[] args) {
        // Create a bank account with $1000 initial balance
        BankAccount account = new BankAccount("John Doe", 1000);

        System.out.println("=== BANK APPLICATION ===\n");

        // Check initial balance
        account.checkBalance();
        System.out.println();

        // Deposit money
        System.out.println("--- Deposit Operation ---");
        account.deposit(500);
    }
}

```

```

        System.out.println();

        // Withdraw money (successful)
        System.out.println("--- Withdraw Operation (Successful) ---");
        try {
            account.withdraw(300);
        } catch (InsufficientFundsException e) {
            System.out.println("Exception: " + e.getMessage());
        }
        System.out.println();

        // Withdraw money (will fail - insufficient funds)
        System.out.println("--- Withdraw Operation (Should Fail) ---");
        try {
            account.withdraw(2000); // This will throw exception
        } catch (InsufficientFundsException e) {
            System.out.println("❌ Transaction Failed!");
            System.out.println("Error Message: " + e.getMessage());
        }
        System.out.println();

        // Final balance
        System.out.println("--- Final Account Status ---");
        account.checkBalance();
    }
}

```

Sample Output

text

=== BANK APPLICATION ===

Account Holder: John Doe

Current Balance: \$1000.0

--- Deposit Operation ---

Deposited: \$500.0

New Balance: \$1500.0

--- Withdraw Operation (Successful) ---

Withdrawn: \$300.0

Remaining Balance: \$1200.0

--- Withdraw Operation (Should Fail) ---

 Transaction Failed!

Error Message: Insufficient funds! Available balance: \$1200.0, Requested: \$2000.0

--- Final Account Status ---

Account Holder: John Doe

Current Balance: \$1200.0